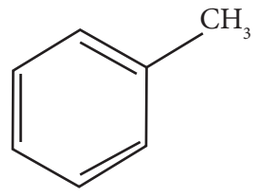
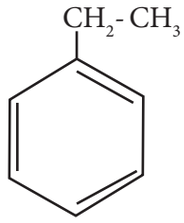
represented by a regular hexagon with a circle inscribed in it.

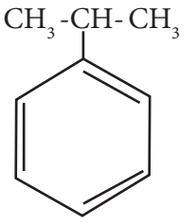
- Since, all the six hydrogen atom in benzene are equivalent, it can give only one mono-substituted compound (Ex) methyl benzene ($C_6H_5$-$CH_3$) which named as toluene.




tolune

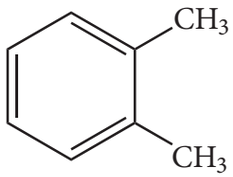



ethylbenzene

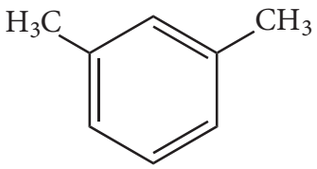



cumene
isopropylbenzene

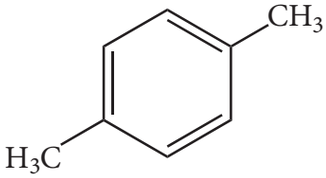
- When di substitution occurs either by a similar monovalent atom or two different atoms or groups in benzene, then three different position isomers are possible. Their relative positions are indicated as ortho (1,2), meta (1,3) and para (1,4). For example, consider dimethyl benzene which is named as xylene.




o-xylene
1,2-dimethylbenzene




m-xylene
1,3-dimethyl benzene




p-xylene
1,4-dimethyl benzene

## Evaluate Yourself

22) write all possible isomers for an aromatic benzenoid compound having the molecular formula $C_8H_{10}$

23) write all possible isomers for a monosubstituted aromatic benzenoid compound having the molecular formula $C_9H_{12}$

### 13.5.2. Aromaticity

Huckel proposed that aromaticity is a function of electronic structure. A compound may be aromatic, if it obeys the following rules

(i) The molecule must be co-planar

(ii) Complete delocalization of π electron in the ring

(iii) Presence of (4n+2) π electrons in the ring where n is an integer (n=0,1,2….)

This is known as Huckel's rule.



Unit 13 - jagan.indd 205 22-01-2021 12:51:38